MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG3DSCBTS201- GENERAL MICROBIOLOGY

Duration: 1.5hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Fill in the blanks
Answer All Questions
Each question carries 1 mark

1._____contributes to negative charge for Gram positive cell**. [U][CO1]**

2 The acid fastness property of *Mycobacteria sp*. is due to the presence of _____ in their cell wall.**[K][CO1]**

3.The ability of a microscope to distinguish between two closely spaced objects is called its _____.**[U][CO1]**

4. Theory of spontaneous generation was disproved by_____.**[K][CO1]**

5.Extracellular infectious virus particles are called _____.**[K] [CO1]**

6.The time required by the bacterial population to double is termed as _____.**[U][CO2]**

7.---------- is a transport medium employed to maintain the viability of Neisseria gonorrhoea**.**

[K][CO2]

8. HEPA stands for _____**[K][CO3]**

9.The process by which bacterial genes are transferred through the agency of bacteriophages is called as _____.**[U][CO5]**

10._____is a type of preservation by which microbes are stored in liquid nitrogen at extremely low temperatures**. [U][CO3]**

[1 x 10= 10]

Part B

Short questions Type Questions

Answer 10 Questions

Each question carries 2 marks

11. Differentiate Protoplast and spheroplast. **[K][CO1]**

12. Define Numerical Taxonomy.**[K] [CO1]**

13. What is Prophage? **[U][CO1]**

14. Comment on Negative staining. **[U][CO1, CO2]**

15.What is the difference between enriched and enrichment media. **[K][CO2]**

16.Give a short note on bacterial growth curve. [**U][CO3]**

17. Comment on Glyoxylate cycle. **[K][CO5]**

18.What are R plasmids?[**K][CO5**]

19.Comment on the principle of Autoclave**.[U][CO3, CO4]**

20.Write a short note on beta lactam antibiotics**. [U] [CO4]**

21.Describe Phenol coefficient test**. [U] [CO3]**

22.Expand IMViC Test**. [U] [CO1, CO2]**

[2 x 10= 20]

Part C

Short Essay Type Questions

Answer five questions

Each question carries 4 marks

23. Give a detailed account on phenetic classification scheme for bacterial taxonomy. **[U] [CO1]**

24. Write an essay on principle and applications of lightMicroscopy. **[U] [CO1]**

25. Describe various cultivation techniques employed in Bacteriology. **[U] [CO2, CO3]**

26. Comment on moist heat sterilization techniques.**[K] [CO3]**

27.Explain the mode of action of antibiotics.**[U] [CO4]**

28.Write a short essay on mode of replication in bacteriophages.**[U] [CO1]**

29.Describe a detailed account on gene transfer methods in bacteria. **[U] [CO5]**

[5 x 4= 20]

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